

Ultimate 32GB Local AI + CKA DevOps Lab Setup Plan

Laptop: Lenovo ThinkPad E14 Gen 3 (Ryzen 5 + Upgraded 32GB RAM)

PHASE 1 – Core Setup

- Install Docker
- Install kind (Kubernetes in Docker)
- Create 3-node local cluster
- Install Ollama for local AI models
- Pull models: llama3:8b, codellama, mistral
- Install Open Interpreter for command execution

PHASE 2 – Kubernetes (CKA) Practice Setup

- Configure kubectl alias (k)
- Enable bash autocomplete
- Practice imperative commands daily
- Simulate exam-style troubleshooting scenarios
- Practice etcd backup & restore
- Practice node upgrade, cordon & drain

PHASE 3 – AI-Powered DevOps Workflow

- Use AI as CKA examiner
- Generate YAML & review configurations
- Automate Ansible playbooks
- Simulate cluster failures & debug
- Run timed mock exams

SYSTEM CAPABILITY AFTER UPGRADE (32GB RAM)

- Run 3-node kind cluster smoothly
- Run 7B–8B LLM models locally
- Use Docker + Kubernetes + AI simultaneously
- No freezing during heavy workloads

CAREER BENEFITS

- Strong CKA readiness
- AI-assisted DevOps workflows
- Platform engineering skill enhancement
- Future-ready Kubernetes + AI automation setup

Recommended RAM Configuration:

2 × 16GB DDR4-3200 SO-DIMM (Dual Channel)